

GARP Risk & AI — Q451–Q550

Unofficial Exam Prep

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Question

A board is defining its AI risk tolerance statement. Which of the following should be included as the most critical element?

- A. A list of every AI model the firm owns.
- B. A clear statement of the types and levels of AI risk the organisation is willing to accept, aligned with strategy and regulatory obligations.
- C. Marketing language about the firm's AI capabilities.
- D. Employee satisfaction metrics.

Answer: B

Explanation

AI risk tolerance defines acceptable risk levels and guides governance. Options A, C, and D are not the core of a risk tolerance statement.

Question

Which statement best distinguishes model risk appetite from model risk tolerance?

- A. They are identical.
- B. Appetite defines the level of risk the organisation actively seeks or accepts to achieve objectives; tolerance defines the maximum deviation from appetite that triggers action.
- C. Appetite is only for technology risk.
- D. Tolerance is only for third-party risk.

Answer: B

Explanation

Appetite = desired risk-return balance; tolerance = acceptable deviation. Options A, C, and D misstate the distinction.

Question

A team trains an LSTM on 20 years of hourly returns to forecast next-hour volatility. The model performs poorly out-of-sample. Which is the most likely reason?

- A. LSTMs cannot handle time-series data.
- B. Financial returns are close to noise at hourly horizons, so complex models overfit non-repeating patterns.
- C. The dataset is too small.
- D. The model should be logistic regression.

Answer: B

Explanation

Financial returns exhibit low signal-to-noise; complex models overfit. Options A, C, and D are incorrect or secondary.

Question

A deep network achieves 99% training accuracy and 62% test accuracy. What action is most appropriate?

- A. Increase model depth.
- B. Apply regularization (dropout, weight decay), reduce complexity, or obtain more data; use early stopping based on validation loss.
- C. Increase learning rate.
- D. Deploy immediately.

Answer: B

Explanation

The large gap signals overfitting. Options A, C, and D would worsen it or ignore the problem.

Question

When preparing data for a distance-based model, which sequence is correct?

- A. Scale, then split into train/test.
- B. Split into train/test, fit scaling on train, apply to test.
- C. Ignore scaling.
- D. Scale only test set.

Answer: B

Explanation

Scaling parameters must be estimated on the training set only, then applied to the test set, to avoid leakage. Options A, C, and D are incorrect.

Question

A model achieves suspiciously high test accuracy. Investigation reveals a feature that is computed using future information. What is the issue and remedy?

- A. Overfitting; add regularization.
- B. Target leakage; the feature must be removed or recomputed using only information available at prediction time.
- C. Underfitting; add features.
- D. Data drift; retrain the model.

Answer: B

Explanation

Future-information features cause leakage. Options A, C, and D do not address the root cause.

Question

A bank wants to use ML to compute regulatory capital. Which of the following is the most critical governance requirement?

- A. Highest possible accuracy.
- B. Full compliance with regulatory standards: documentation, validation, interpretability, and independent review by qualified validators.
- C. Fastest training.
- D. Lowest cost.

Answer: B

Explanation

Regulatory reporting requires compliance, documentation, validation, and interpretability. Options A, C, and D are insufficient.

Question

A firm uses a vendor-supplied AI model for credit scoring. Which governance gap is most likely?

- A. The model cannot be tested.
- B. Limited transparency into model internals, restricting full independent validation; mitigation includes black-box testing and contractual rights to documentation.
- C. Vendor models cannot be used in regulated industries.
- D. Vendor models have no risk.

Answer: B

Explanation

Vendor models limit transparency; black-box validation and contract terms help. Options A, C, and D are inaccurate.

Question

Which of the following is essential for reproducibility of an ML model?

- A. Only the model architecture.
- B. Fixed random seeds, documented data versions, environment specifications, and code versioning.
- C. Only training data.
- D. Only hyperparameters.

Answer: B

Explanation

Reproducibility requires fixed seeds, data versions, environment, and code versioning. Options A, C, and D are partial.

Question

Why is data versioning important in AI governance?

- A. To reduce storage.
- B. To enable reproducibility, auditability, and root-cause analysis of model behaviour over time.
- C. To speed up training.
- D. To reduce costs.

Answer: B

Explanation

Data versioning supports reproducibility and audit. Options A, C, and D are secondary.

Question

What is the purpose of the K-means++ initialization?

- A. To randomly select centroids.
- B. To spread initial centroids far apart, reducing the chance of poor local optima and improving convergence.
- C. To eliminate outliers.
- D. To reduce K.

Answer: B

Explanation

K-means++ chooses centroids with probability proportional to squared distance, spreading them out. Options A, C, and D misstate the purpose.

Question

A dataset has clusters with very different densities. Which algorithm is likely to struggle?

- A. K-means.
- B. DBSCAN, because it uses a single global density threshold.
- C. Hierarchical clustering.
- D. Gaussian Mixture Models.

Answer: B

Explanation

DBSCAN struggles with varying densities because a single ϵ cannot fit all clusters. Options A, C, and D are less affected.

Question

Which linkage method is most sensitive to outliers?

- A. Single linkage.
- B. Complete linkage.
- C. Average linkage.
- D. Ward's method.

Answer: A

Explanation

Single linkage uses the closest points and can chain clusters via single points, including outliers. Options B, C, and D are more robust.

Question

What is a common limitation of the elbow method for choosing K?

- A. It always produces a clear elbow.
- B. The elbow may be ambiguous or absent, and its identification is subjective.
- C. It only works for $K=2$.
- D. It requires labels.

Answer: B

Explanation

Elbow plots may not have a clear bend. Options A, C, and D are inaccurate.

Question

Why is the silhouette method often used alongside the elbow method?

- A. Because both methods always agree.
- B. Because silhouette scores provide a quantitative criterion (higher is better) that complements the visual elbow heuristic.
- C. Because silhouette requires labels.
- D. Because elbow is never used.

Answer: B

Explanation

Silhouette provides a numerical criterion to complement visual inspection. Options A, C, and D are inaccurate.

Question

A random forest reports that a feature "region_id" is highly important. Why should this be interpreted with caution?

- A. Feature importance is always wrong.
- B. High-cardinality features can appear artificially important due to the impurity bias; permutation importance may give a more reliable ranking.
- C. Region has no business meaning.
- D. Only tree depth matters.

Answer: B

Explanation

Impurity-based importance is biased toward high-cardinality features. Options A, C, and D are less accurate.

Question

Which kernel is most appropriate when the data is not linearly separable and the analyst has no prior knowledge of the boundary shape?

- A. Linear kernel.
- B. RBF kernel.
- C. Polynomial of very high degree.
- D. No kernel.

Answer: B

Explanation

The RBF kernel is a common default for non-linear separation. Options A, C, and D are less suitable defaults.

Question

Which statement best describes the curse of dimensionality for KNN?

- A. KNN gets faster as dimensionality increases.
- B. As dimensionality increases, distances become less discriminative and all points appear equidistant, degrading KNN performance.
- C. KNN is unaffected by dimensionality.
- D. KNN performs best in very high dimensions.

Answer: B

Explanation

High dimensions cause distance concentration. Options A, C, and D are incorrect.

Question

Increasing the regularization strength in a linear model will:

- A. Decrease bias and increase variance.
- B. Increase bias and decrease variance, potentially reducing overfitting.
- C. Have no effect.
- D. Increase both bias and variance.

Answer: B

Explanation

Regularization trades bias for variance reduction. Options A, C, and D misstate the trade-off.

Question

Why is cross-validation preferable to a single validation split for hyperparameter tuning?

- A. It is faster.
- B. It reduces the variance of performance estimates and uses data more efficiently.
- C. It requires more data.
- D. It eliminates the need for a test set.

Answer: B

Explanation

Cross-validation reduces variance of performance estimates. Options A, C, and D are incorrect or incomplete.

Question

A pipeline fails to correctly identify negations. Which preprocessing step is most likely missing?

- A. Tokenization.
- B. N-grams or negation handling — bi-grams and above capture “not good” as a unit.
- C. Lowercasing.
- D. Feature extraction.

Answer: B

Explanation

N-grams preserve local order and handle negation. Options A, C, and D do not address negation.

Question

A word appears in every document of a corpus. What is its TF-IDF score?

- A. Very high.
- B. Zero (or near-zero), because $IDF = \log(N/N) = 0$.
- C. Undefined.
- D. Equal to TF.

Answer: B

Explanation

$IDF = \log(N/n)$ where n = document frequency. If $n=N$, $IDF=0$. Options A, C, and D are wrong.

Question

A classic limitation of static word embeddings (Word2Vec) is:

- A. They are too large.
- B. They assign a single vector per word, ignoring polysemy (multiple meanings).
- C. They cannot handle text.
- D. They always overfit.

Answer: B

Explanation

Static embeddings assign one vector per word, missing context-dependent meanings. Options A, C, and D are incorrect.

Question

Contextual embeddings from transformers differ from static embeddings by:

- A. Having fewer dimensions.
- B. Producing different vectors for the same word depending on the surrounding context.
- C. Being faster.
- D. Using no training data.

Answer: B

Explanation

Contextual embeddings depend on surrounding tokens; static embeddings do not. Options A, C, and D are incorrect.

Question

Which best explains why transformers replaced RNNs in most NLP tasks?

- A. Transformers use less memory.
- B. Transformers parallelise over sequence positions and capture long-range dependencies through attention.
- C. Transformers use fewer parameters.
- D. Transformers avoid training.

Answer: B

Explanation

Transformers' parallelism and attention capabilities are the key advantages. Options A, C, and D are incorrect or secondary.

Question

Setting the temperature of an LLM to a high value will:

- A. Make outputs more deterministic.
- B. Increase randomness and creativity, at the cost of predictability.
- C. Decrease token count.
- D. Have no effect.

Answer: B

Explanation

High temperature flattens the token distribution, increasing randomness. Options A, C, and D are incorrect.

Question

Top-P (nucleus) sampling works by:

- A. Selecting all tokens with probability above a fixed threshold.
- B. Selecting the smallest set of tokens whose cumulative probability exceeds P.
- C. Selecting the K most probable tokens.
- D. Selecting tokens at random.

Answer: B

Explanation

Top-P is cumulative-probability-based. Options A, C, and D describe other methods.

Question

Why do LLMs hallucinate?

- A. They are trained on too little data.
- B. They optimise for statistical plausibility, not factual accuracy, generating fluent but incorrect outputs.
- C. They have no memory.
- D. They require more tokens.

Answer: B

Explanation

Hallucinations stem from training objectives that optimise plausibility. Options A, C, and D are unrelated.

Question

RAG improves LLM factual accuracy by:

- A. Fine-tuning the model.
- B. Retrieving relevant documents at inference time and providing them as context.
- C. Increasing model size.
- D. Reducing context length.

Answer: B

Explanation

RAG retrieves external knowledge and provides it to the model. Options A, C, and D do not describe RAG.

Question

Which is the most effective prompt engineering technique for structured output?

- A. Short random prompts.
- B. Specifying role, format, and providing a few-shot example.
- C. Ignoring output format.
- D. Reducing context.

Answer: B

Explanation

Role specification, format guidance, and examples reduce ambiguity. Options A, C, and D are ineffective.

Question

Chain-of-thought prompting improves performance mainly because:

- A. It changes model weights.
- B. It allows the model to use intermediate reasoning tokens to solve multi-step problems.
- C. It reduces computation.
- D. It increases randomness.

Answer: B

Explanation

Chain-of-thought provides intermediate reasoning steps. Options A, C, and D are incorrect.

Question

A key caveat of relying on LLM leaderboards is:

- A. They are always definitive.
- B. Benchmark contamination and saturation can inflate or compress scores, and leaderboards may not reflect real-world performance.
- C. They ignore accuracy.
- D. They cover all tasks.

Answer: B

Explanation

Benchmark issues include contamination, saturation, and lack of real-world relevance. Options A, C, and D are incorrect.

Question

What does “stateless” mean for an LLM?

- A. It stores all previous conversations.
- B. It does not retain information between separate prompts unless prior context is re-supplied.
- C. It cannot process sequences.
- D. It works only offline.

Answer: B

Explanation

LLMs process each prompt independently. Options A, C, and D are incorrect.

Question

What distinguishes an agentic AI system from a standard LLM application?

- A. Higher accuracy.
- B. The ability to autonomously take actions in an environment with limited supervision.
- C. Lower cost.
- D. Smaller model size.

Answer: B

Explanation

Agentic AI takes autonomous actions. Options A, C, and D are not the distinguishing feature.

Question

Why might a GenAI model score highly on a benchmark yet perform poorly in production?

- A. Benchmarks are always accurate.
- B. Benchmarks may be contaminated, saturated, or unrepresentative of the real-world context in which the model is deployed.
- C. Benchmarks require no data.
- D. Benchmarks measure only speed.

Answer: B

Explanation

Benchmarks have known limitations. Options A, C, and D are incorrect.

Question

Which scenario best favours RAG over fine-tuning?

- A. When proprietary knowledge changes frequently and must be updated often.
- B. When model weights must change permanently.
- C. When data is static.
- D. When latency is irrelevant.

Answer: A

Explanation

RAG is best for frequently changing knowledge. Options B, C, and D are less aligned with RAG's strengths.

Question

Which is an appropriate method for evaluating bias in a GenAI system?

- A. Ignore bias.
- B. Use demographic-parity-like metrics, stereotype scores, and adversarial prompts across groups.
- C. Only evaluate accuracy.
- D. Only evaluate speed.

Answer: B

Explanation

Bias evaluation requires fairness metrics and adversarial testing. Options A, C, and D are insufficient.

Question

A model card for a GenAI system typically includes:

- A. Only the model's API key.
- B. Intended use, training data, evaluation results, limitations, and ethical considerations.
- C. Only training cost.
- D. Only temperature settings.

Answer: B

Explanation

Model cards document intended use, training data, evaluations, limitations, and ethics. Options A, C, and D are partial.

Question

Which of the following is a significant deployment risk for GenAI systems?

- A. Uncontrolled output that could produce non-compliant or harmful content.
- B. Faster inference.
- C. Reduced storage.
- D. Lower cost.

Answer: A

Explanation

Uncontrolled output is a key deployment risk. Options B, C, and D are benefits, not risks.

Question

Prompt injection is:

- A. A harmless input.
- B. A manipulation where an attacker inserts malicious instructions into prompts or data to cause unintended behaviour.
- C. A model architecture.
- D. A training technique.

Answer: B

Explanation

Prompt injection attacks manipulate the model via crafted inputs. Options A, C, and D misstate.

Question

Which is an example of individual fairness?

- A. Equal approval rates across groups.
- B. Two applicants with identical qualifications receive the same decision.
- C. Equal TPR across groups.
- D. Equal FPR across groups.

Answer: B

Explanation

Individual fairness = similar individuals treated similarly. Options A, C, and D are group fairness metrics.

Question

The impossibility result for fairness holds:

- A. Only when base rates are equal.
- B. When base rates differ, making it impossible to satisfy demographic parity, predictive rate parity, and equal opportunity simultaneously.
- C. Never.
- D. Only for regression models.

Answer: B

Explanation

The impossibility arises under unequal base rates. Options A, C, and D are incorrect.

Question

Which is most likely to cause data composition bias?

- A. Balanced dataset.
- B. Imbalanced dataset where minority class is underrepresented.
- C. Random data.
- D. Complete data.

Answer: B

Explanation

Class imbalance drives composition bias. Options A, C, and D do not.

Question

Which is an example of measurement bias?

- A. Inconsistent measurement methods across groups.
- B. Random sampling.
- C. Balanced data.
- D. Complete data.

Answer: A

Explanation

Measurement bias arises when features are measured differently across groups. Options B, C, and D are unrelated.

Question

Which is an example of historical bias?

- A. Training on past hiring decisions that reflect discrimination against women.
- B. Random sampling.
- C. Balanced data.
- D. Normalized features.

Answer: A

Explanation

Historical bias stems from past discrimination encoded in data. Options B, C, and D are unrelated.

Question

Which is an example of proxy discrimination?

- A. Explicit use of race as a feature.
- B. Use of a feature highly correlated with race (e.g., certain ZIP codes).
- C. Removing race.
- D. Balanced data.

Answer: B

Explanation

Proxies carry discriminatory effects indirectly. Options A, C, and D are different or unrelated.

Question

Which is true about a logistic regression model?

- A. It is interpretable but often not explainable for complex cases.
- B. It is inherently interpretable and generally explainable.
- C. Neither interpretable nor explainable.
- D. Only explainable.

Answer: B

Explanation

Logistic regression has interpretable coefficients and can be explained. Options A, C, and D are inaccurate.

Question

Which technique is most commonly used to explain black-box model predictions?

- A. Linear regression.
- B. SHAP or LIME.
- C. Logistic regression.
- D. Grid search.

Answer: B

Explanation

SHAP and LIME are post-hoc explainability techniques. Options A, C, and D are unrelated.

Question

The interpretability-accuracy trade-off implies:

- A. Interpretable models are always less accurate.
- B. Complex models often achieve higher accuracy but lower interpretability; the trade-off should be managed based on context.
- C. Complex models are always better.
- D. Interpretability is irrelevant.

Answer: B

Explanation

The trade-off is real and should be managed contextually. Options A, C, and D are oversimplifications.

Question

Which of the following is a direct source of AI-related reputational risk?

- A. Increased storage.
- B. Privacy breaches, algorithmic bias, or failure to explain decisions.
- C. Faster training.
- D. Higher accuracy.

Answer: B

Explanation

Reputational risk arises from privacy, bias, and explainability failures. Options A, C, and D are not sources of reputational risk.

Question

Which is a clear violation of human autonomy?

- A. Explaining a decision.
- B. Manipulating a user's choices through dark patterns or AI-driven persuasion without consent.
- C. Providing choices.
- D. Showing accurate information.

Answer: B

Explanation

Manipulative design undermines autonomy. Options A, C, and D respect autonomy.

Question

Which of the following is a societal risk?

- A. A bank loses money.
- B. An AI system amplifies political misinformation at scale.
- C. A chatbot mislabels a customer.
- D. A model underperforms.

Answer: B

Explanation

Societal risk affects society at large. Options A, C, and D are individual or organisational risks.

Question

Which is a key concern in AI existential risk discussions?

- A. Model speed.
- B. Superintelligent AI misaligned with human values.
- C. Storage costs.
- D. IT complexity.

Answer: B

Explanation

Existential risk focuses on superintelligence and value alignment. Options A, C, and D are operational.

Question

AI may worsen global inequality because:

- A. AI is universally accessible.
- B. Advanced AI capability concentrated in a few nations may widen economic and political gaps.
- C. AI reduces inequality.
- D. AI has no impact.

Answer: B

Explanation

Concentration of AI can widen gaps. Options A, C, and D are inaccurate.

Question

Why is GenAI particularly concerning for misinformation campaigns?

- A. It cannot generate text.
- B. It makes realistic content generation cheap and scalable.
- C. It only produces truthful content.
- D. It is too slow.

Answer: B

Explanation

Generative AI makes content creation cheap and scalable. Options A, C, and D are incorrect.

Question

AI's ability to infer sensitive attributes from data creates which risk?

- A. Faster computation.
- B. Privacy and surveillance risk.
- C. Lower storage costs.
- D. Higher accuracy.

Answer: B

Explanation

Inference capabilities create privacy risks. Options A, C, and D are unrelated.

Question

Which of the following is a category of GenAI risk?

- A. Functional risks.
- B. Data and information risks.
- C. Organizational and strategic risks.
- D. All of the above.

Answer: D

Explanation

GenAI risks span functional, data, organizational, strategic, and societal categories. All three are included.

Question

Which is a data risk unique to GenAI?

- A. Storage cost.
- B. Training data may include proprietary, personal, or copyrighted material, creating legal exposure.
- C. Training time.
- D. Model size.

Answer: B

Explanation

Copyright, privacy, and proprietary data in training corpora create legal exposure. Options A, C, and D are operational.

Question

Which is an organizational risk of GenAI?

- A. Responsibility gaps when multiple teams deploy GenAI without clear ownership.
- B. Faster inference.
- C. Lower cost.
- D. Reduced storage.

Answer: A

Explanation

Responsibility gaps are a key organizational risk. Options B, C, and D are benefits.

Question

Which is a strategic risk of GenAI?

- A. Vendor lock-in and dependence on a small number of providers.
- B. Faster inference.
- C. Lower cost.
- D. Reduced storage.

Answer: A

Explanation

Vendor lock-in is a strategic risk. Options B, C, and D are benefits.

Question

Which statement best captures utilitarianism in AI ethics?

- A. It judges actions by adherence to rules.
- B. It judges actions by their consequences, maximising overall welfare.
- C. It focuses on character.
- D. It ignores outcomes.

Answer: B

Explanation

Utilitarianism judges actions by consequences and aggregate welfare. Options A, C, and D are other frameworks or the opposite.

Question

Which statement best captures deontology in AI ethics?

- A. Actions are judged by consequences.
- B. Actions are judged by adherence to duties and rules regardless of consequences.
- C. Focus on character.
- D. Focus on profit.

Answer: B

Explanation

Deontology judges by duties and rules. Options A, C, and D are different frameworks.

Question

Which statement best captures virtue ethics?

- A. Focus on consequences.
- B. Focus on duties.
- C. Focus on character traits and living a good life.
- D. Focus on laws.

Answer: C

Explanation

Virtue ethics focuses on character. Options A, B, and D are different or incomplete.

Question

Why can different concepts of justice conflict?

- A. They are identical.
- B. Egalitarianism, utilitarianism, and meritocracy prioritise different values, which can yield conflicting allocations.
- C. Only utilitarianism matters.
- D. Justice is unambiguous.

Answer: B

Explanation

Different justice concepts conflict. Options A, C, and D misstate.

Question

Which is essential for respecting autonomy in AI systems?

- A. Ignoring user preferences.
- B. Obtaining informed consent and avoiding manipulative designs.
- C. Hiding decisions.
- D. Coercion.

Answer: B

Explanation

Respecting autonomy requires consent and non-manipulation. Options A, C, and D violate autonomy.

Question

Which is an example of applying nonmaleficence in AI?

- A. Deploying untested models.
- B. Testing AI thoroughly and removing unjustified risks before deployment.
- C. Ignoring safety.
- D. Racing to deploy.

Answer: B

Explanation

Nonmaleficence requires avoiding harm, including unnecessary risk. Options A, C, and D violate the principle.

Question

Which is an example of beneficence in AI?

- A. Avoiding harm only.
- B. Actively designing AI to improve welfare, e.g., improving access to credit for underserved populations.
- C. Ignoring user needs.
- D. Maximising profit only.

Answer: B

Explanation

Beneficence = actively doing good. Options A, C, and D are inconsistent with beneficence.

Question

Which of the following is a privacy best practice?

- A. Data minimisation and purpose limitation.
- B. Unlimited retention.
- C. Sharing with third parties without consent.
- D. Unencrypted storage.

Answer: A

Explanation

Minimisation and purpose limitation are core privacy practices. Options B, C, and D violate them.

Question

Which is NOT a GDPR data subject right?

- A. Access.
- B. Erasure.
- C. Portability.
- D. Right to sell data.

Answer: D

Explanation

GDPR grants access, erasure, portability, rectification, restriction, and objection. Selling data is not a right.

Question

Which is explicitly prohibited under the EU AI Act?

- A. Fraud detection.
- B. Social scoring and manipulative AI.
- C. Credit scoring.
- D. Customer service chatbots.

Answer: B

Explanation

Social scoring and manipulative AI are prohibited. Other uses are regulated but not prohibited.

Question

Which is NOT a benefit of an AI ethics framework?

- A. Provides guidelines for ethical dilemmas.
- B. Builds trust.
- C. Reduces regulatory risk.
- D. Guarantees model accuracy.

Answer: D

Explanation

Ethics frameworks do not guarantee model accuracy. Options A, B, and C are recognised benefits.

Question

The primary purpose of an AI ethics committee is to:

- A. Write code.
- B. Review high-risk use cases and provide guidance on ethical dilemmas.
- C. Market AI products.
- D. Train models.

Answer: B

Explanation

Ethics committees review use cases and give guidance. Options A, C, and D are operational or unrelated.

Question

Which is an example of a US state-level privacy law?

- A. GDPR.
- B. CCPA (California).
- C. PIPL (China).
- D. DSA (EU).

Answer: B

Explanation

CCPA is California's privacy law. Options A, C, and D are non-US or different jurisdictions.

Question

GDPR fines can reach:

- A. €1,000.
- B. €100,000.
- C. Up to €20 million or 4% of global annual turnover.
- D. No fines.

Answer: C

Explanation

GDPR fines reach €20M or 4% of global turnover. Options A, B, and D understate the maximum.

Question

The EU AI Act's maximum fines reach:

- A. ≤1 million.
- B. ≤5 million.
- C. Up to ≤35 million or 7% of global turnover.
- D. No fines.

Answer: C

Explanation

The AI Act allows fines up to ≤35M or 7% of global turnover. Options A, B, and D are incorrect.

Question

Which is a core Responsible AI principle?

- A. Maximise profit only.
- B. Fairness, transparency, accountability, and privacy.
- C. Ignore stakeholders.
- D. Hide AI use.

Answer: B

Explanation

Responsible AI includes fairness, transparency, accountability, and privacy. Options A, C, and D are contrary.

Question

Which is an example of global explainability?

- A. SHAP for a single prediction.
- B. Feature importance across the full model.
- C. Counterfactual for one instance.
- D. LIME for one case.

Answer: B

Explanation

Feature importance is a global technique. Options A, C, and D are local.

Question

In a regulated credit decision context, explainability requires:

- A. Publishing all model internals.
- B. Meaningful justifications for individual decisions, typically via SHAP, counterfactuals, or adverse action notices.
- C. Only aggregate metrics.
- D. Refusing to explain.

Answer: B

Explanation

Explainability requires meaningful per-decision justification. Options A, C, and D are insufficient or wrong.

Question

Which is a group fairness metric?

- A. Individual fairness.
- B. Demographic parity.
- C. Consistency.
- D. Predictability.

Answer: B

Explanation

Demographic parity is a group fairness metric. Options A, C, and D are different or unrelated.

Question

Why can't all fairness metrics be satisfied simultaneously under unequal base rates?

- A. Because fairness is subjective.
- B. Because mathematical constraints make demographic parity, predictive rate parity, and equal opportunity mutually incompatible.
- C. Because GDPR forbids it.
- D. Because of computational limits.

Answer: B

Explanation

The impossibility result is mathematical. Options A, C, and D misstate the cause.

Question

Which best describes the data governance committee's role?

- A. Operationally process data.
- B. Provide oversight on policies, standards, and practices, and resolve escalations.
- C. Build models.
- D. Market data.

Answer: B

Explanation

The committee provides oversight and resolves escalations. Options A, C, and D are operational roles.

Question

Which activity is part of data quality management?

- A. Only storage.
- B. Data profiling, cleansing, validation, and monitoring.
- C. Only encryption.
- D. Only backup.

Answer: B

Explanation

Data quality management includes profiling, cleansing, validation, and monitoring. Options A, C, and D are related but partial.

Question

Data lineage documents:

- A. Only the file size.
- B. The origin, movement, and transformations of data across systems.
- C. Only the encryption status.
- D. Only the storage location.

Answer: B

Explanation

Lineage tracks origin, movement, and transformations. Options A, C, and D are partial.

Question

A metadata repository typically stores:

- A. Raw data.
- B. Definitions, relationships, schemas, and technical/business metadata.
- C. Marketing content.
- D. Only encryption keys.

Answer: B

Explanation

Metadata repositories store definitions, schemas, and relationships. Options A, C, and D are unrelated or partial.

Question

Which is a standard data classification level?

- A. Public.
- B. Internal.
- C. Restricted.
- D. All of the above.

Answer: D

Explanation

Standard classification: public, internal, confidential, restricted. All three are typical levels.

Question

The principle of least privilege requires:

- A. Granting maximum access.
- B. Granting users only the access necessary for their role.
- C. No access.
- D. Full open access.

Answer: B

Explanation

Least privilege = only what is needed. Options A, C, and D are inconsistent.

Question

Which is a widely adopted standard for encrypting data at rest?

- A. MD5.
- B. AES-256.
- C. SHA-1.
- D. Base64.

Answer: B

Explanation

AES-256 is a standard symmetric encryption algorithm. Options A, C, and D are not encryption standards for confidentiality.

Question

Which practice should be part of maintaining a model inventory?

- A. Update the inventory only annually.
- B. Update on each material change, including new models, retirements, and changes in risk tier.
- C. Never update.
- D. Only capture model names.

Answer: B

Explanation

The inventory is updated on material changes. Options A, C, and D are insufficient.

Question

Which factor most raises the risk tier of an AI model?

- A. Small user base.
- B. High materiality, high complexity, or high exposure.
- C. Short documentation.
- D. Low cost.

Answer: B

Explanation

Risk tiering considers materiality, complexity, and exposure. Options A, C, and D do not raise risk.

Question

Which activity is part of independent model validation?

- A. Rebuilding the model.
- B. Assessing conceptual soundness, data quality, implementation, and performance.
- C. Marketing the model.
- D. Entering data.

Answer: B

Explanation

Validation covers conceptual soundness, data, implementation, and performance. Options A, C, and D are not validation activities.

Question

A model validation report typically includes:

- A. Validation opinion, findings, limitations, and recommendations.
- B. Only the model's accuracy.
- C. Only the code.
- D. Only the training data.

Answer: A

Explanation

Validation reports include opinion, findings, limitations, and recommendations. Options B, C, and D are partial.

Question

Which is typically included in a model risk appetite statement?

- A. A specific number of models.
- B. The firm's tolerance for model risk and escalation criteria.
- C. Marketing material.
- D. Only costs.

Answer: B

Explanation

Appetite statements define tolerance and escalation criteria. Options A, C, and D are unrelated.

Question

A model risk committee typically includes:

- A. Only developers.
- B. Senior representatives from risk, business, validation, and possibly audit.
- C. Only IT.
- D. Only external consultants.

Answer: B

Explanation

Cross-functional membership ensures diverse oversight. Options A, C, and D are incomplete.

Question

Model risk reporting should be:

- A. Occasional.
- B. Regular and ad hoc, with appropriate escalation to the board.
- C. Only annual.
- D. Only when problems occur.

Answer: B

Explanation

Reporting is regular and ad hoc. Options A, C, and D are insufficient.

Question

Which is a hallmark of a strong model risk culture?

- A. Silence about issues.
- B. Open communication, accountability, and continuous learning.
- C. Ignoring concerns.
- D. Only senior leaders caring.

Answer: B

Explanation

A strong culture values openness, accountability, and learning. Options A, C, and D are contrary.

Question

Model risk training should cover:

- A. Only technical content.
- B. Technical, governance, ethical, and regulatory aspects.
- C. Only ethics.
- D. Only regulation.

Answer: B

Explanation

Comprehensive training covers all four areas. Options A, C, and D are too narrow.

Question

A material model change should trigger:

- A. Immediate deployment.
- B. Impact assessment, documentation, validation, and approval.
- C. No action.
- D. Retirement.

Answer: B

Explanation

Material changes require structured review. Options A, C, and D violate governance.

Question

Model retirement requires:

- A. Silent deletion.
- B. Formal documentation, inventory update, and archival.
- C. Continued use.
- D. No governance involvement.

Answer: B

Explanation

Retirement is formal and documented. Options A, C, and D violate governance.

Question

Responsible AI governance requires:

- A. Only technical controls.
- B. A combination of technical controls, governance policies, and cultural elements.
- C. Only policies.
- D. Only culture.

Answer: B

Explanation

Responsible AI governance integrates technical, policy, and cultural elements. Options A, C, and D are insufficient.

Question

The most important guiding principle in AI risk and governance is:

- A. Maximise model complexity.
- B. Balance performance with governance, validation, monitoring, fairness, explainability, and ethical alignment with regulatory and business expectations.
- C. Ignore risks.
- D. Deploy all models without review.

Answer: B

Explanation

Responsible AI balances performance and governance. Options A, C, and D contradict the principle.